IECE 672 - Foundations of Statistical Inference Project Report

Joy Saha

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1 Part 1

Dataset summary

The dataset contains $N = 529$ citizen-reported service requests in Albany County (submitted through the SeeClickFix platform) labelled with one of four service categories: Parking Enforcement (139), Code Violations (156), Signs — Missing, Needed, or Damaged (124), and Traffic Signal Repairs (110). Each request is described by a feature vector $\mathbf{x} = (x_1, \ldots, x_m)$ with $m = 2{,}535$, where x_i is the number of times the ith preprocessed word appears in the request title or description.

1.1 Task 1 — Binary Hypothesis Testing under the Bayes Criterion

Two macro-classes for the binary test (Task 1). Following the assignment, we group the four service categories into two classes:

- **Class (a)** = Parking Enforcement ∪ Code Violations (295 requests)
- **Class (b)** = Signs (Missing, Needed, or Damaged) ∪ Traffic Signal Repairs (234 requests)

The binary hypothesis test is then H_0 : class (a) vs. H_1 : class (b). Task 2 returns to all four categories as separate hypotheses.

Train / test split. As specified, half of the requests in each of the four *original* categories are placed in the training set and the rest in the test set. This stratifies the split inside each macro-class, avoiding any imbalance between Parking and Code (within class (a)) or between Signs and Traffic (within class (b)). The resulting split has 264 training and 265 test requests; the same partition is used for both likelihood models and for both tasks.

(i) Approach: derivations and key equations

Bayes-optimal binary decision rule. Let H_0, H_1 have priors $P_0, P_1 = 1 - P_0$, and let C_{ij} be the cost of deciding H_i when H_j is true. The Bayes risk is

$$\mathcal{R}(C_{00}, C_{01}, C_{10}, C_{11}) \;=\; \sum_{j=0}^{1} P_j \sum_{i=0}^{1} C_{ij} \;\Pr(\text{decide } H_i \mid H_j). \tag{1}$$

Assuming $C_{10} > C_{00}$ and $C_{01} > C_{11}$, pointwise minimisation of (1) yields the likelihood-ratio test (LRT)

$$\Lambda(\mathbf{x}) \;\triangleq\; \frac{P(\mathbf{x} \mid H_1)}{P(\mathbf{x} \mid H_0)} \;\underset{H_0}{\overset{H_1}{\gtrless}}\; \eta, \qquad \boxed{\eta(C_{00}, C_{01}, C_{10}, C_{11}) \;=\; \frac{(C_{10} - C_{00})\, P_0}{(C_{01} - C_{11})\, P_1}.} \tag{2}$$

Equivalently, writing the threshold as $\gamma \triangleq \eta$ (we keep this name throughout, working in the LR domain), the LRT becomes

$$\Lambda(\mathbf{x}) \underset{H_0}{\overset{H_1}{\gtrless}} \gamma, \qquad \text{or, equivalently in the log domain,} \qquad \log\Lambda(\mathbf{x}) \underset{H_0}{\overset{H_1}{\gtrless}} \log\gamma. \tag{3}$$

We evaluate (3) in the log domain to avoid underflow over $m = 2{,}535$ features.

Functional dependence of the metrics on the costs. The four performance metrics are conditional probabilities of the LRT outcome:

$$P_F(\gamma) \;=\; \sum_{\mathbf{x}\,:\,\Lambda(\mathbf{x})>\gamma} P(\mathbf{x} \mid H_0), \tag{4}$$

$$P_D(\gamma) \;=\; \sum_{\mathbf{x}\,:\,\Lambda(\mathbf{x})>\gamma} P(\mathbf{x} \mid H_1), \tag{5}$$

$$P_M(\gamma) \;=\; \sum_{\mathbf{x}\,:\,\Lambda(\mathbf{x})<\gamma} P(\mathbf{x} \mid H_1) \;=\; 1 - P_D(\gamma), \tag{6}$$

$$P_\epsilon(\gamma) \;=\; P_0\, P_F(\gamma) + P_1\, P_M(\gamma). \tag{7}$$

For evaluation we replace these probabilities by their empirical (test-set) counterparts. Let $\mathcal{T}_0, \mathcal{T}_1$ denote the test requests truly belonging to H_0 and H_1. Then

$$\widehat{P}_F(\gamma) \;=\; \frac{1}{|\mathcal{T}_0|} \sum_{\mathbf{x}\in\mathcal{T}_0\,:\,\Lambda(\mathbf{x})\,>\,\gamma} 1, \tag{8}$$

$$\widehat{P}_D(\gamma) \;=\; \frac{1}{|\mathcal{T}_1|} \sum_{\mathbf{x}\in\mathcal{T}_1\,:\,\Lambda(\mathbf{x})\,>\,\gamma} 1, \tag{9}$$

$$\widehat{P}_M(\gamma) \;=\; \frac{1}{|\mathcal{T}_1|} \sum_{\mathbf{x}\in\mathcal{T}_1\,:\,\Lambda(\mathbf{x})\,<\,\gamma} 1, \tag{10}$$

$$\widehat{P}_\epsilon(\gamma) \;=\; \widehat{P}_0\, \widehat{P}_F(\gamma) + \widehat{P}_1\, \widehat{P}_M(\gamma). \tag{11}$$

The decision threshold γ depends on the cost tuple via (2).

Two structural facts simplify the table:

1. The cost tuple enters (8)–(11) only through the threshold γ in (2); the priors P_0, P_1 are fixed by training.
2. By (2), γ depends on the costs only through the ratio $r = (C_{10} - C_{00})/(C_{01} - C_{11})$. Without loss of generality we may set $C_{00} = C_{11} = 0$ (no cost for correct decisions, the standard convention), so the metrics depend on the full four-tuple only through this single ratio.

Accordingly, we adopt the convention that correct decisions cost zero ($C_{00} = C_{11} = 0$) and report $P_F, P_D, P_M, P_\epsilon$ over a grid of ten misclassification cost pairs (C_{01}, C_{10}) that varies the ratio C_{10}/C_{01} from 1/10 (heavy missed-detection penalty, decisions favour H_1) through 1 (uniform 0/1 costs, MAP rule) to 10 (heavy false-alarm penalty, decisions favour H_0). For the ROC, we instead sweep γ directly over a dense grid from $\gamma \to 0$ to $\gamma = 100$, which traces the curve from the "always declare H_1" corner ($\gamma \to 0$) toward the "never declare H_1" corner ($\gamma \to \infty$).

Multi-variate Bernoulli model. This model treats every feature x_i as a binary value: $x_i = 1$ if word i appears in the request, $x_i = 0$ otherwise. Under feature independence,

$$P(\mathbf{x} \mid H_j) \;=\; \prod_{i=1}^{m} P(x_i \mid H_j)^{b} \left(1 - P(x_i \mid H_j)\right)^{1-b}, \qquad b \in \{0,1\},\ j = 0,1, \tag{12}$$

exactly eq. (1) of the assignment, with $b \equiv x_i$ the realised binary value of the ith feature. The prior is estimated by $\widehat{P}_j = N_j/N$ (N_j = number of training requests in H_j, $N = N_0 + N_1$); the class-conditional probabilities by the smoothed maximum-likelihood estimate

$$\widehat{P}(x_i \mid H_j) \;=\; \frac{N_{ij} + 1}{N_j + 2}, \tag{13}$$

which is eq. (2) of the assignment, where N_{ij} is the number of training requests in class j that contain feature x_i.

Multinomial model. Using raw term frequencies, class-conditional probabilities are estimated as

$$\widehat{P}(x_i \mid H_j) \;=\; \frac{tf_{ij} + 1}{N_j + V}, \tag{14}$$

which is eq. (3) of the assignment, where tf_{ij} is the sum of raw term frequencies of word x_i across all training requests in class j, N_j is the sum of all term frequencies in class j, and $V = 2{,}535$ is the vocabulary size. The conditional density is

$$P(\mathbf{x} \mid H_j) \;=\; \prod_{i=1}^{m} \widehat{P}(x_i \mid H_j), \tag{15}$$

which is eq. (4) of the assignment. The multinomial coefficient $\binom{\sum_i x_i}{x_1,\dots,x_m}$ is the same under H_0 and H_1, so it cancels in the LRT (3) and is dropped throughout.

Numerically stable LLR. Stacking the test feature matrix $\mathbf{X} \in \mathbb{R}^{n \times V}$, both log-likelihoods are linear in $\mathbf{X}$:

$$\log P_{\text{Bern}}(\mathbf{x} \mid H_j) = \mathbf{b}^{\top} \log \widehat{\mathbf{p}}_j + (\mathbf{1} - \mathbf{b})^{\top} \log(\mathbf{1} - \widehat{\mathbf{p}}_j), \qquad \mathbf{b}_i = \nVdash\{x_i > 0\}, \tag{16}$$

$$\log P_{\text{Mult}}(\mathbf{x} \mid H_j) = \mathbf{x}^{\top} \log \widehat{\mathbf{p}}_j. \tag{17}$$

(ii) Evaluation procedure

1. Map the four service categories to the two macro-classes: class (a) $\to H_0$, class (b) $\to H_1$.
2. Stratified 50/50 split inside each *original* category, so half of Parking, Code, Signs, and Traffic each go to training, the rest to testing.
3. Estimate $\widehat{P}_j$ and $\widehat{P}(x_i \mid H_j)$ from the training set only via (13) or (14).
4. For every test request, compute the likelihood ratio $\Lambda(\mathbf{x}) = P(\mathbf{x} \mid H_1)/P(\mathbf{x} \mid H_0)$ from (3) (we evaluate it in the log domain to avoid underflow).
5. For each cost tuple $(C_{00}, C_{01}, C_{10}, C_{11})$ in the cost grid (Tables 1–2), compute the threshold γ via (2) and tabulate $P_F, P_D, P_M, P_\epsilon$ via (8)–(11).
6. Trace the ROC by sweeping γ over a dense grid from 0 to 100 (covering the full range of the LR distribution on this dataset), and compute the AUC by the trapezoidal rule.

(iii) Results and discussion

Setup. Estimated priors from the training half: $\widehat{P}_0 = 147/264 = 0.557$ for class (a), $\widehat{P}_1 = 117/264 = 0.443$ for class (b). Training size = 264; test size = 265.

Table 1: Multi-variate Bernoulli model: $P_F, P_D, P_M, P_\epsilon$ as a function of $(C_{00}, C_{01}, C_{10}, C_{11})$ for class (a) vs. class (b) (AUC = 0.9907).

C_{00}	C_{01}	C_{10}	C_{11}	γ	P_F	P_D	P_M	P_ϵ
0	1	1	0	1.256	0.007	0.949	0.051	0.026
0	1	2	0	2.513	0.007	0.932	0.068	0.034
0	1	5	0	6.282	0.007	0.915	0.085	0.042
0	1	10	0	12.564	0.000	0.915	0.085	0.038
0	2	1	0	0.628	0.007	0.949	0.051	0.026
0	5	1	0	0.251	0.007	0.966	0.034	0.019
0	10	1	0	0.126	0.007	0.974	0.026	0.015
0	3	2	0	0.838	0.007	0.949	0.051	0.026
0	2	3	0	1.885	0.007	0.932	0.068	0.034
0	4	5	0	1.571	0.007	0.932	0.068	0.034

Table 2: Multinomial model: $P_F, P_D, P_M, P_\epsilon$ as a function of $(C_{00}, C_{01}, C_{10}, C_{11})$ for class (a) vs. class (b) (AUC = 0.9866).

C_{00}	C_{01}	C_{10}	C_{11}	γ	P_F	P_D	P_M	P_ϵ
0	1	1	0	1.256	0.034	0.966	0.034	0.034
0	1	2	0	2.513	0.034	0.966	0.034	0.034
0	1	5	0	6.282	0.027	0.966	0.034	0.030
0	1	10	0	12.564	0.014	0.966	0.034	0.023
0	2	1	0	0.628	0.047	0.966	0.034	0.042
0	5	1	0	0.251	0.054	0.974	0.026	0.042
0	10	1	0	0.126	0.054	0.983	0.017	0.038
0	3	2	0	0.838	0.047	0.966	0.034	0.042
0	2	3	0	1.885	0.034	0.966	0.034	0.034
0	4	5	0	1.571	0.034	0.966	0.034	0.034

Discussion. The two macro-classes are highly separable: AUC = 0.9907 for the Bernoulli model and 0.9866 for the multinomial, and the uniform-cost test errors are $\widehat{P}_\epsilon = 0.026$ (Bernoulli) and 0.034 (multinomial), corresponding to 7 and 9 misclassifications out of 265 test requests respectively.

How the metrics depend on the costs. The threshold γ varies smoothly with the cost ratio C_{10}/C_{01} (compare the γ columns), but the test-set metrics are step functions of γ because there are only finitely many test points to cross. This produces the following observations in the tables:

- Different cost tuples can produce the same operating point. For example, in Table 1 the tuples $(0, 1, 1, 0)$, $(0, 2, 1, 0)$, and $(0, 3, 2, 0)$ all yield $(P_F, P_D) = (0.007, 0.949)$ because no test-set LR values lie between their respective γ's.

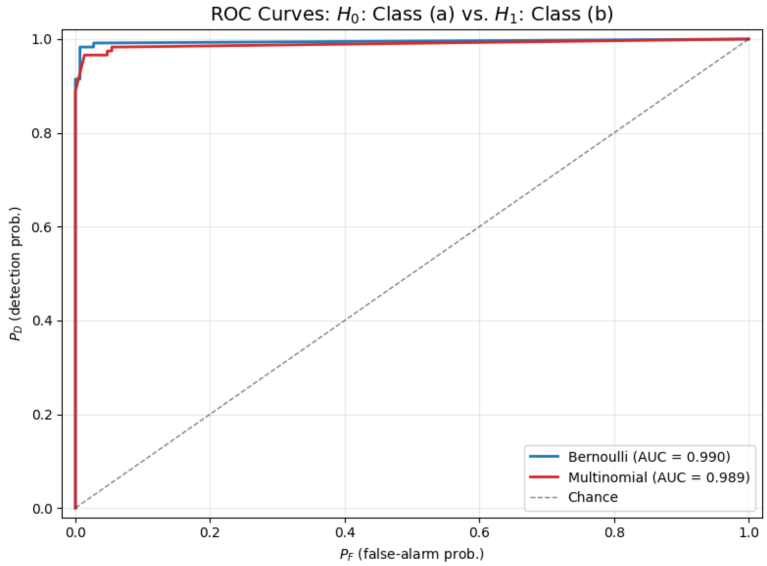


Figure 1: ROC for class (a) (H_0) vs. class (b) (H_1), traced by sweeping the LR threshold γ over a dense grid from 0 to 100. Markers show selected γ values; the curve enters at the upper-right corner ($\gamma \to 0$, "always declare H_1") and exits at the origin ($\gamma \to \infty$, "never declare H_1").

- As C_{10}/C_{01} increases (false-alarm penalty grows), γ rises and P_F weakly decreases while P_M weakly increases. In Table 2 P_F steps down from 0.054 at $\gamma = 0.126$ all the way to 0.014 at $\gamma = 12.564$; Table 1 shows the same pattern but P_F is already nearly zero throughout this range so the variation is hidden.
- As C_{10}/C_{01} decreases (missed-detection penalty grows), γ falls and P_D weakly increases. The Bernoulli model improves P_D from 0.949 at $\gamma = 1.256$ (uniform costs) to 0.974 at $\gamma = 0.126$; the multinomial improves P_D from 0.966 to 0.983 over the same range.
- P_ϵ in (7) is the uniform-cost Bayes error, so the cost tuple affects it only by selecting which operating point we sit at on the ROC.

ROC observations. Both ROCs in Fig. 1 hug the upper-left corner, confirming near-perfect separability. The Bernoulli ROC dominates the multinomial ROC slightly. As γ decreases past 1, the curve climbs steeply toward the corner; for $\gamma \geq 2$ both curves are essentially flat at very small P_F, indicating that H_0 and H_1 have very few overlapping LR values. Sweeping γ all the way to 100 does not move the operating point beyond what is already reached at $\gamma \approx 10$.

Why both models do well. Class (a) (parking and code-related issues) and class (b) (signage and traffic-signal issues) use largely disjoint vocabularies (e.g. "meter", "car", "building", "trash" in (a) vs. "sign", "light", "signal", "pole" in (b)), so the class-conditional word distributions are well separated. Most words in the corpus appear at most once per request — the median non-zero count is exactly 1 — so the binarised Bernoulli likelihood and the count-based multinomial likelihood agree on the bulk of the evidence. The small Bernoulli advantage in AUC is consistent with the fact that with so few token repetitions per request, the multinomial likelihood gains little extra information from raw counts but is more sensitive to mass-imbalanced documents.

1.2 Task 2 — Multiple Hypothesis Testing under the Bayes Criterion

(i) Approach: derivations and key equations

Hypotheses and decision rule. We now classify each test request among all four service categories. Index the hypotheses as

- H_0: Parking Enforcement
- H_1: Code Violations
- H_2: Signs (Missing, Needed, or Damaged)
- H_3: Traffic Signal Repairs

with priors $P_0, \ldots, P_3$ and cost matrix $\{C_{ij}\}_{i,j=0}^{3}$. Following the assignment, instead of the full M-ary MAP rule we treat H_0 as a baseline and run three pairwise binary LRTs of H_k against H_0 ($k = 1, 2, 3$). The Bayes-optimal threshold of each pairwise test follows from (2):

$$\Lambda_1(\mathbf{x}) \triangleq \frac{P(\mathbf{x} \mid H_1)}{P(\mathbf{x} \mid H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \frac{P_0(C_{10} - C_{00})}{P_1(C_{01} - C_{11})} = \gamma_1, \tag{18}$$

$$\Lambda_2(\mathbf{x}) \triangleq \frac{P(\mathbf{x} \mid H_2)}{P(\mathbf{x} \mid H_0)} \underset{H_0}{\overset{H_2}{\gtrless}} \frac{P_0(C_{20} - C_{00})}{P_2(C_{02} - C_{22})} = \gamma_2, \tag{19}$$

$$\Lambda_3(\mathbf{x}) \triangleq \frac{P(\mathbf{x} \mid H_3)}{P(\mathbf{x} \mid H_0)} \underset{H_0}{\overset{H_3}{\gtrless}} \frac{P_0(C_{30} - C_{00})}{P_3(C_{03} - C_{33})} = \gamma_3. \tag{20}$$

For simplicity we set $\gamma_1 = \gamma_2 = \gamma_3 = \gamma^\star$, with $\gamma^\star \in \mathbb{Z}_+$. The classifier declares H_k ($k \in \{1, 2, 3\}$) iff the corresponding $\Lambda_k(\mathbf{x}) > \gamma^\star$; otherwise it declares H_0. When the test is run at the natural operating point $\gamma^\star \approx 1$ (uniform costs, MAP rule), the resulting four-way assignments form a confusion matrix that we report below.

Functional dependence of the metrics on $\gamma^\star$. With the three pairwise tests sharing the same threshold, the four performance metrics generalise (4)–(7) to averages of three terms:

$$P_F(\gamma^\star) = \frac{1}{3}\left[\sum_{\mathbf{x}:\Lambda_1(\mathbf{x})>\gamma^\star} P(\mathbf{x} \mid H_0) + \sum_{\mathbf{x}:\Lambda_2(\mathbf{x})>\gamma^\star} P(\mathbf{x} \mid H_0) + \sum_{\mathbf{x}:\Lambda_3(\mathbf{x})>\gamma^\star} P(\mathbf{x} \mid H_0)\right], \tag{21}$$

$$P_D(\gamma^\star) = \frac{1}{3}\left[\sum_{\mathbf{x}:\Lambda_1(\mathbf{x})>\gamma^\star} P(\mathbf{x} \mid H_1) + \sum_{\mathbf{x}:\Lambda_2(\mathbf{x})>\gamma^\star} P(\mathbf{x} \mid H_2) + \sum_{\mathbf{x}:\Lambda_3(\mathbf{x})>\gamma^\star} P(\mathbf{x} \mid H_3)\right], \tag{22}$$

$$P_M(\gamma^\star) = \frac{1}{3}\left[\sum_{\mathbf{x}:\Lambda_1(\mathbf{x})<\gamma^\star} P(\mathbf{x} \mid H_1) + \sum_{\mathbf{x}:\Lambda_2(\mathbf{x})<\gamma^\star} P(\mathbf{x} \mid H_2) + \sum_{\mathbf{x}:\Lambda_3(\mathbf{x})<\gamma^\star} P(\mathbf{x} \mid H_3)\right], \tag{23}$$

$$P_e(\gamma^\star) = P(e \mid H_0)\,P_0 + P(e \mid H_1)\,P_1 + P(e \mid H_2)\,P_2 + P(e \mid H_3)\,P_3, \tag{24}$$

where the per-hypothesis error probabilities are

$$P(e \mid H_0) = P_F(\gamma^\star), \tag{25}$$

$$P(e \mid H_k) = \sum_{\mathbf{x}:\Lambda_k(\mathbf{x})<\gamma^\star} P(\mathbf{x} \mid H_k), \qquad k = 1, 2, 3. \tag{26}$$

For evaluation, the population sums are replaced by their empirical (test-set) counterparts: each sum becomes a fraction of the corresponding test sub-sample for which the displayed inequality holds, exactly as in (8)–(11).

Multi-variate Bernoulli model. The likelihoods are still given by (12), but with $M = 4$ classes the smoothed maximum-likelihood estimate of the class-conditional word probabilities becomes

$$\widehat{P}(x_i \mid H_j) \;=\; \frac{N_{ij}+1}{N_j+4}, \qquad j = 0,1,2,3, \tag{27}$$

where N_{ij} is the number of training requests in class H_j that contain feature x_i, and N_j is the total number of training requests in class H_j. Priors are estimated by

$$\widehat{P}_j \;=\; \frac{N_j}{N_0+N_1+N_2+N_3}, \qquad j = 0,1,2,3. \tag{28}$$

Multinomial model. The same likelihood form (15) and smoothed estimate (14) apply, summed over the four classes; only the number of priors changes, again given by (28).

Cost tuple in the M = 4 case. For the cost-sweep table we use the same convention as in Task 1 ($C_{00} = C_{kk} = 0$, $k = 1,2,3$, i.e. correct classification costs zero), with a single off-diagonal cost $C_{0k} \equiv C_{01}$ for "deciding H_0 when truth is H_k" and a single $C_{k0} \equiv C_{10}$ for "deciding H_k when truth is H_0". Substituting into (18)–(20) and equating the three thresholds gives a shared $\gamma^\star$ that depends on the four-tuple via

$$\gamma^\star(C_{00}, C_{01}, C_{10}, C_{11}) \;=\; \frac{(C_{10}-C_{00})\,P_0}{(C_{01}-C_{11})\,\bar{P}_{1:3}}, \qquad \bar{P}_{1:3} \;\triangleq\; \frac{P_1+P_2+P_3}{3}, \tag{29}$$

where $\bar{P}_{1:3}$ is the average prior of the three non-baseline hypotheses (this normalisation is consistent with the metrics in (21)–(23), which themselves average over the three branches).

(ii) Evaluation procedure

1. Stratified 50/50 split inside each of the four categories, so half of Parking, Code, Signs, and Traffic each go to training and the rest to testing (264 train, 265 test).
2. Estimate priors $\widehat{P}_j$ via (28) and class-conditional word probabilities $\widehat{P}(x_i \mid H_j)$ via (27) or (14) from the training set only.
3. For every test request, compute the three pairwise log-likelihood ratios $\log \Lambda_k(\mathbf{x}) = \log P(\mathbf{x} \mid H_k) - \log P(\mathbf{x} \mid H_0)$ for $k = 1,2,3$ in the log domain.
4. For each cost tuple $(C_{00}, C_{01}, C_{10}, C_{11})$ in the cost grid, compute the shared threshold $\gamma^\star$ via (29) and tabulate P_F, P_D, P_M, P_e via the empirical counterparts of (21)–(24).
5. At the natural operating point $\gamma^\star \approx 1$, classify every test request and form the 4×4 confusion matrix together with the per-class one-vs-rest detection metrics.
6. Trace the ROC by sweeping $\gamma^\star$ over a dense grid covering $[0, 100]$ and compute the AUC by the trapezoidal rule.

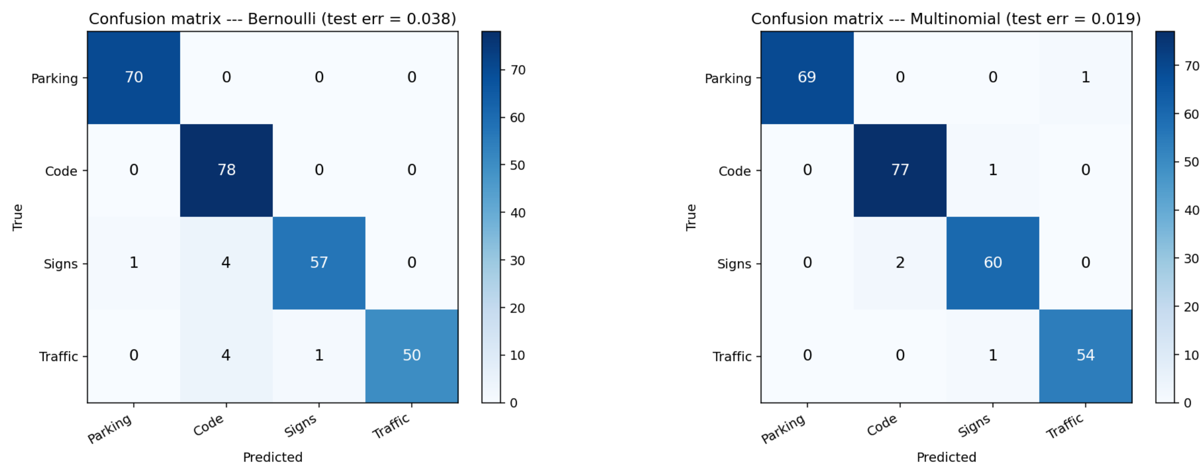
(iii) Results and discussion

Setup. Estimated priors (training half):

$$\widehat{P}_0 = 0.261\ (\mathrm{H}_0),\quad \widehat{P}_1 = 0.295\ (\mathrm{H}_1),\quad \widehat{P}_2 = 0.235\ (\mathrm{H}_2),\quad \widehat{P}_3 = 0.208\ (\mathrm{H}_3),$$

so $\bar{P}_{1:3} = 0.246$. Training size $= 264$; test size $= 265$.

Confusion matrices and per-class metrics at $\gamma^\star \approx 1$. At the uniform-cost operating point, the Bernoulli model misclassifies 10/265 test requests (overall test error 0.038) and the multinomial model misclassifies 5/265 (overall test error 0.019). The full 4×4 confusion matrices are shown in Fig. 2; per-class one-vs-rest detection metrics are summarised in Table 3.



(a) Bernoulli model.

(b) Multinomial model.

Figure 2: Confusion matrices at $\gamma^\star \approx 1$ (rows = true class, columns = predicted class). Bernoulli: 10/265 errors; multinomial: 5/265 errors.

Table 3: Per-class one-vs-rest detection metrics at $\gamma^\star \approx 1$. "$P_D^{(k)}$" is the fraction of true-H_k requests classified as H_k; "$P_F^{(k)}$" is the fraction of non-H_k requests classified as H_k; $P_M^{(k)} = 1 - P_D^{(k)}$.

Class	Bernoulli			Multinomial		
	$P_D^{(k)}$	$P_F^{(k)}$	$P_M^{(k)}$	$P_D^{(k)}$	$P_F^{(k)}$	$P_M^{(k)}$
Parking Enforcement	1.000	0.005	0.000	0.986	0.000	0.014
Code Violations	1.000	0.043	0.000	0.987	0.011	0.013
Signs (Missing, Needed, or Damaged)	0.919	0.005	0.081	0.968	0.010	0.032
Traffic Signal Repairs	0.909	0.000	0.091	0.982	0.005	0.018

Cost-sweep tables. Tables 4–5 report the four metrics as a function of $(C_{00}, C_{01}, C_{10}, C_{11})$, with the shared threshold $\gamma^\star$ computed from (29).

Table 4: Multi-variate Bernoulli model: P_F, P_D, P_M, P_e as a function of $(C_{00}, C_{01}, C_{10}, C_{11})$ for the 4-hypothesis test (AUC $= 0.9916$).

C_{00}	C_{01}	C_{10}	C_{11}	$\gamma^\star$	P_F	P_D	P_M	P_e
0	1	1	0	1.062	0.000	0.972	0.028	0.019
0	1	2	0	2.123	0.000	0.961	0.039	0.027
0	1	5	0	5.308	0.000	0.961	0.039	0.027
0	1	10	0	10.615	0.000	0.961	0.039	0.027
0	2	1	0	0.531	0.000	0.972	0.028	0.019
0	5	1	0	0.212	0.000	0.978	0.022	0.015
0	10	1	0	0.106	0.010	0.978	0.022	0.018
0	3	2	0	0.708	0.000	0.972	0.028	0.019
0	2	3	0	1.592	0.000	0.967	0.033	0.023
0	4	5	0	1.327	0.000	0.972	0.028	0.019

Table 5: Multinomial model: P_F, P_D, P_M, P_e as a function of $(C_{00}, C_{01}, C_{10}, C_{11})$ for the 4-hypothesis test (AUC $= 0.9951$).

C_{00}	C_{01}	C_{10}	C_{11}	$\gamma^\star$	P_F	P_D	P_M	P_e
0	1	1	0	1.062	0.005	0.989	0.011	0.009
0	1	2	0	2.123	0.005	0.979	0.021	0.016
0	1	5	0	5.308	0.005	0.969	0.031	0.024
0	1	10	0	10.615	0.005	0.969	0.031	0.024
0	2	1	0	0.531	0.010	0.995	0.005	0.006
0	5	1	0	0.212	0.010	0.995	0.005	0.006
0	10	1	0	0.106	0.014	1.000	0.000	0.004
0	3	2	0	0.708	0.010	0.989	0.011	0.010
0	2	3	0	1.592	0.005	0.989	0.011	0.009
0	4	5	0	1.327	0.005	0.989	0.011	0.009

Discussion. Both models perform very well in the 4-hypothesis setting: AUC $= 0.9916$ for Bernoulli and 0.9951 for multinomial. At the natural operating point $\gamma^\star \approx 1$ (uniform-cost regime), the Bernoulli model achieves $(P_F, P_D, P_M, P_e) = (0.000, 0.972, 0.028, 0.019)$ and the multinomial achieves $(0.005, 0.989, 0.011, 0.009)$.

Confusion-matrix observations (Fig. 2). The Bernoulli model's 10 errors split as: 1 Signs request misclassified as Parking, 4 Signs requests misclassified as Code, and 4 Traffic Signal requests misclassified as Code (plus 1 Traffic misclassified as Signs). All errors fall on classes that share vocabulary with Code Violations, which is the largest training class and therefore the hardest baseline to dislodge. The multinomial model reduces this to 5 errors: 1 Parking $\rightarrow$ Traffic, 1 Code $\rightarrow$ Signs, 2 Signs $\rightarrow$ Code, 1 Traffic $\rightarrow$ Signs. Counting word repetitions (instead of just presence/absence) helps disambiguate Signs vs. Traffic — the two categories that most often share words like “signal”, “light”, “pole” — because the multinomial likelihood is sensitive to how often those words appear in a single request.

Per-class behaviour (Table 3). Bernoulli is perfect on the two largest classes ($P_D^{(k)} = 1.000$ for Parking and Code) but loses ground on the two smallest classes (Signs $P_D = 0.919$, Traffic

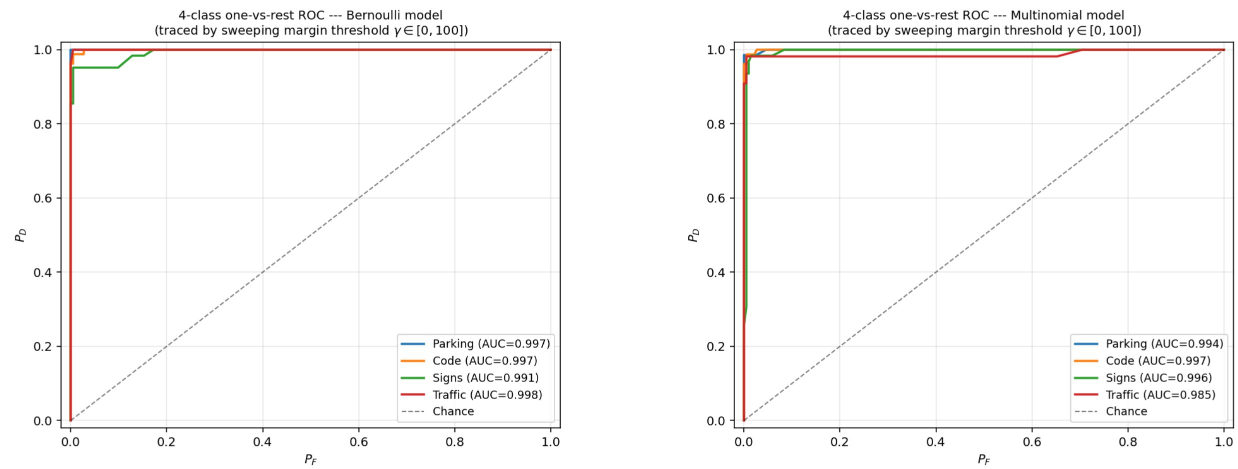


Figure 3: ROC for the 4-hypothesis test, traced by sweeping the shared threshold $\gamma^\star$ over a dense grid in $[0, 100]$. Left: multi-variate Bernoulli ($\text{AUC} = 0.9916$). Right: multinomial ($\text{AUC} = 0.9951$). Red circles label selected $\gamma^\star$ values; green squares mark the cost-grid operating points from Tables 4–5.

$P_D = 0.909$); their training samples are smallest (62 and 55 in the training half), so Laplace smoothing has a relatively larger effect on the estimated word probabilities. The multinomial model trades a small amount of detection on the two largest classes ($P_D = 0.986, 0.987$) for substantially better detection on the two smallest ($P_D = 0.968, 0.982$) — a worthwhile trade because the smaller classes were the bottleneck for the overall error.

How the metrics depend on the costs. As in Task 1, the cost tuple enters the metrics only through the threshold $\gamma^\star$ in (29), and the test-set metrics are step functions of $\gamma^\star$:

- As C_{10}/C_{01} increases (false-alarm penalty grows), $\gamma^\star$ rises and the test becomes more conservative about declaring any of H_1, H_2, H_3. In Table 5, P_F stays at 0.005 across rows 1–4 while P_D steps down from 0.989 to 0.969; P_M and P_e rise correspondingly.
- As C_{10}/C_{01} decreases (missed-detection penalty grows), $\gamma^\star$ falls and the test declares the non-baseline hypotheses more readily. The multinomial P_D rises to 1.000 at $\gamma^\star = 0.106$ (row 7), at the modest cost of $P_F = 0.014$. The Bernoulli model is already near-saturated on P_D (≥ 0.972) for any $\gamma^\star \leq 1$, so further reductions of $\gamma^\star$ buy little detection improvement on this dataset.
- Different cost tuples can produce nearly identical operating points: for example, $(0, 3, 2, 0)$, $(0, 4, 5, 0)$ and $(0, 2, 1, 0)$ all yield very close metric values in Table 4 because their $\gamma^\star$ values fall in the same "flat" region of the LR distribution.
- P_e in (24) reaches its minimum near $\gamma^\star \approx 0.1$–0.5 on this dataset (multinomial $P_e = 0.004$ at $\gamma^\star = 0.106$), confirming that for these priors and word distributions the optimal Bayes test sits slightly below the equal-cost MAP point.

ROC observations. Both ROCs in Fig. 3 hug the upper-left corner. The multinomial curve dominates the Bernoulli curve on this task — the multinomial likelihood gains from the small amount of within-document repetition that distinguishes the categories. Across all cost tuples in the heavy-FA-penalty regime ($C_{10} \in \{2, 5, 10\}$), P_F for both models is at most 0.005, indicating that the test almost never confuses Parking Enforcement with one of the other three classes regardless of how the threshold is set.

Comparison to Task 1. In Task 1 the binary test between class (a) and class (b) achieved AUC ≈ 0.99 with a uniform-cost test error of 2.6–3.4%. Going to the four-hypothesis test in Task 2, the metric definitions change (averages over three pairwise tests instead of a single binary test), but the operating-point behaviour is qualitatively the same: ROCs hug the upper-left corner and the test error at $\gamma^\star \approx 1$ is comparable ($P_e = 0.009$–0.019). The four service categories partition the topic space into largely non-overlapping word distributions, so the multi-hypothesis generalisation inherits the favourable separation already observed in the binary task. The hypothesis test could be improved further by allowing per-branch thresholds $\gamma_1, \gamma_2, \gamma_3$ tuned individually rather than using the simplifying assumption $\gamma_1 = \gamma_2 = \gamma_3$.

Conclusion

We designed Bayes-optimal hypothesis tests for the SeeClickFix Albany County dataset under both the multi-variate Bernoulli (eqs. (1)–(2) of the assignment) and multinomial (eqs. (3)–(4)) likelihood models with maximum-likelihood (Laplace-smoothed) parameter estimates. Task 1 implemented a binary test between the two macro-classes (a) and (b); both models achieved AUC above 0.98 (Bernoulli 0.9907, multinomial 0.9866) with uniform-cost test errors of 2.6% and 3.4% respectively. Task 2 implemented a 4-hypothesis test using three pairwise LRTs of H_1, H_2, H_3 against H_0 with a shared threshold $\gamma^\star$; AUCs rose to 0.9916 (Bernoulli) and 0.9951 (multinomial), with overall test errors at $\gamma^\star \approx 1$ of 3.8% (10/265) and 1.9% (5/265) respectively. The confusion matrices show that residual errors concentrate on Signs and Traffic Signal Repairs in the Bernoulli model, while the multinomial model spreads its few errors more evenly across categories. In both tasks, sweeping the cost tuple $(C_{00}, C_{01}, C_{10}, C_{11})$ shifts the threshold exactly as predicted by the Bayes formula, and sweeping the threshold itself from 0 to 100 traces the full ROC. The hypothesis test in Task 2 could be improved further by tuning the three pairwise thresholds independently instead of imposing $\gamma_1 = \gamma_2 = \gamma_3$.